

Touhidul Alam Seyam

Software Engineer | Full-Stack Developer | Applied AI Researcher

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PROFESSIONAL SUMMARY

Software engineer, full-stack developer, and applied AI researcher who turns technical ideas into production-ready products and reproducible experiments. Professional experience since 2021 across TypeScript, Python, React, Next.js, Node.js, backend services, data systems, and machine learning. Builds client applications, open-source developer tools, AI agents, and research software; author or co-author of ten source-backed research works.

CORE SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Go, Zig

Product Engineering: React, Next.js, Node.js, Django, FastAPI, REST APIs, React Native, Tailwind CSS

Data and Infrastructure: PostgreSQL, MongoDB, Redis, SQLite, Prisma, Convex, Docker, Git, AWS, Vercel

AI and Research: scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, computer vision, agent evaluation, reproducible experiments

PROFESSIONAL EXPERIENCE

Software Engineer | Hello World Communications Ltd | Chattogram, Bangladesh | Aug 2024-Present

- Build, review, deploy, and maintain full-stack web applications for client and internal use across frontend, backend, database, and AI-integration layers.
- Collaborate with cross-functional stakeholders on requirements, implementation, production releases, and ongoing application support.

Freelance Developer | Self-employed | Remote | Mar 2021-2025

- Delivered web applications from requirements and interface design through backend implementation, deployment, and ongoing support.
- Built digital products for small businesses, nonprofit and academic organizations, and event teams using modern JavaScript and Python stacks.

Research Assistant | BGC Trust University Bangladesh | Chattogram, Bangladesh | Jul 2023-2025

- Supported faculty-led machine-learning research through experiment design, data analysis, implementation, academic writing, and publication preparation.
- Contributed to work spanning agricultural computer vision, healthcare risk modeling, malware classification, LLM performance, and high-performance clustering.

SELECTED PROJECTS

bun-scikit | TypeScript, Bun, Zig | github.com/Seyamalam/bun-scikit

- Built a scikit-learn-inspired machine-learning library with native Zig acceleration, tests and benchmark gates, 209 tracked runtime exports, and documented APIs across preprocessing, model selection, ensembles, clustering, and metrics.

RoboFusion | Python, FastAPI, WebSockets, ESP32 | github.com/Seyamalam/Robofusion

- Developed a multi-hazard campus response platform with device integration, authentication, persistence, incident-lifecycle management, real-time dashboards, automated tests, and load-testing tools.

ASRRO Portal | Next.js, TypeScript, Convex, Better Auth | github.com/Seyamalam/asrro

- Developed a public website and role-aware organization-management system covering membership approvals, events, attendance, content, reports, notifications, and restricted finance workflows.

Kaggriculture Autonomous Agent | Python, Simulation, Evaluation | github.com/Seyamalam/Kaggriculture

- Engineered a deterministic farming agent and evaluation harness with seeded seat-swapped tournaments, frozen replay corpora, manifest hashing, and regression gates.

EDUCATION

B.Sc. (Hons.) in Computer Science and Engineering | BGC Trust University Bangladesh | Chattogram, Bangladesh | Expected Dec 2026

Focus: software engineering, machine learning, computer vision, applied AI, and research-driven product development.

SELECTED RESEARCH

- “Architectures for AI-Driven Visual Assistance: Evaluating Server-Mediated Mobile and Direct Access Desktop Client Implementations.” Lecture Notes in Networks and Systems, 2026. DOI: [10.1007/978-3-032-15764-5_45](https://doi.org/10.1007/978-3-032-15764-5_45)
- “AgriScan: Next.js powered cross-platform solution for automated plant disease diagnosis and crop health management.” Journal of Electrical Systems and Information Technology, 2024. DOI: [10.1186/s43067-024-00169-7](https://doi.org/10.1186/s43067-024-00169-7)
- “Next-Generation K-Means Clustering: Mojo-Driven Performance for Big Data.” International Journal of Intelligent Information Systems, 2025. DOI: [10.11648/j.ijjis.20251401.12](https://doi.org/10.11648/j.ijjis.20251401.12)

SELECTED CERTIFICATIONS

IBM Applied Data Science with Python; IBM Deep Learning with TensorFlow; IEEE Computer Society Machine Learning Mastery; Cisco Python Essentials 1 and 2; Harvard CS50 for Educators